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Digital Image Fundamentals and Computer Vision

ITA0506 Computer Vision

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Introduction

- **What is a Digital Image?**
- A **digital image** is a visual representation of a real-world scene stored as a collection of tiny picture elements called **pixels**.
- **Key Components**
- **Pixel:** Smallest element of an image
- **Resolution:** Total number of pixels (e.g., 1920×1080)
- **Gray-scale Image:** Contains only intensity values
- **RGB Image:** Uses Red, Green, and Blue colour channels
- **Why is it Important?**
- Stores visual information digitally
- Enables image processing and analysis
- Forms the foundation of Computer Vision and Artificial Intelligence



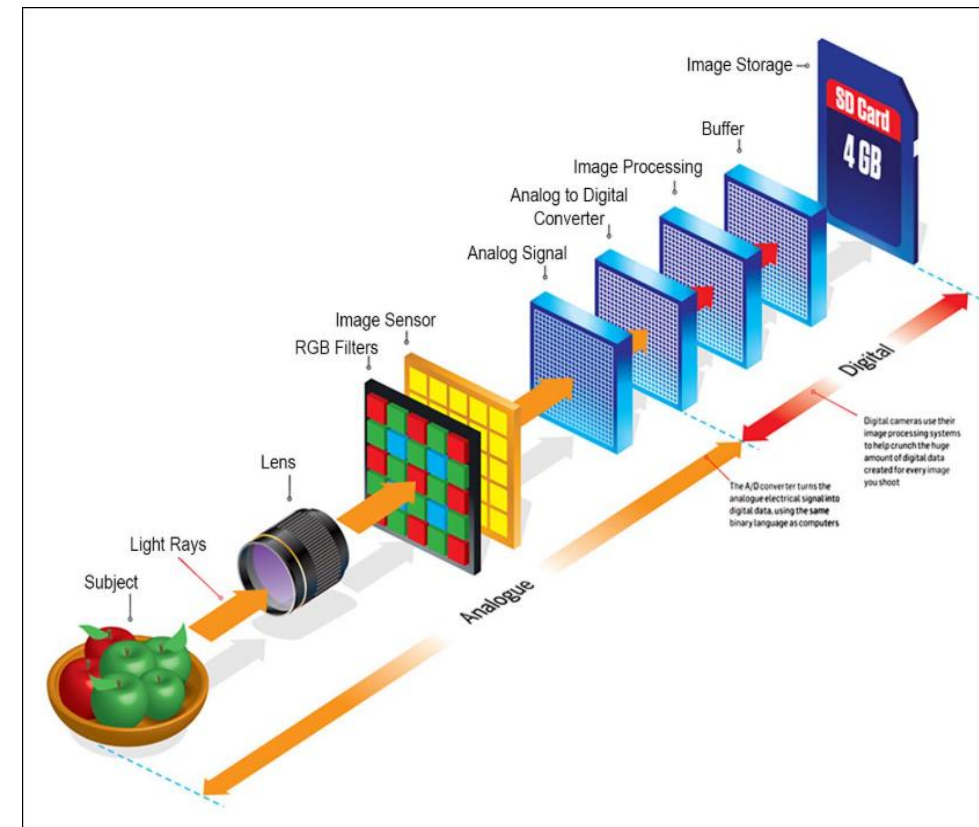
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Image Acquisition & Image Formation

- **Image Acquisition**
- **Definition**
- Image acquisition is the process of capturing a real-world scene using an imaging device and converting it into a digital image.
- **Process**
- Light falls on the object.
- The object reflects light.
- The camera lens focuses the light.
- The image sensor captures the focused image.





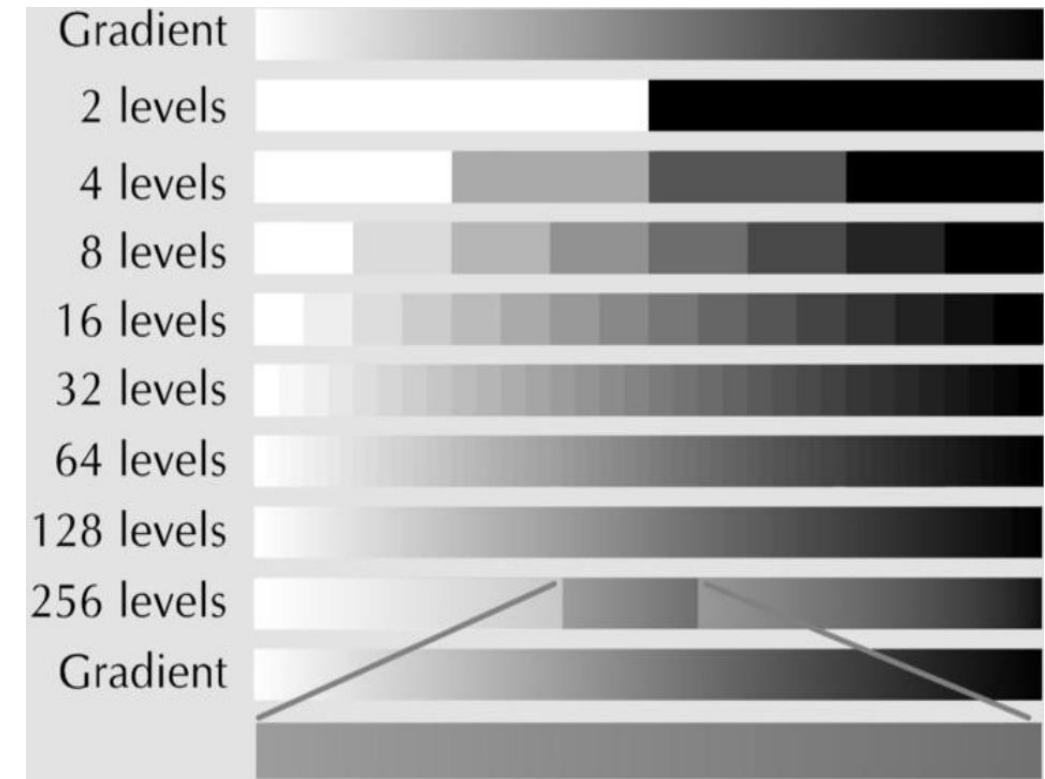
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Sampling and Quantization

- **Sampling**
- **Definition**
- Sampling converts a continuous image into a discrete grid of pixels.
- **Purpose**
- Determines **spatial resolution**.
- Controls the amount of image detail.
- Higher sampling → Better image quality.
- Lower sampling → Pixelated image.





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Quantization

- **Definition**
- Quantization is the process of converting continuous pixel intensity values into a finite number of discrete digital levels. It assigns each pixel to the nearest available intensity level, making the image suitable for digital storage and processing.
- **How Quantization Works**
- The image sensor captures continuous intensity values.
- These values are rounded to the nearest digital level.
- Each pixel is assigned a binary value based on the selected bit depth.
- Higher bit depth provides more intensity levels and smoother image quality.

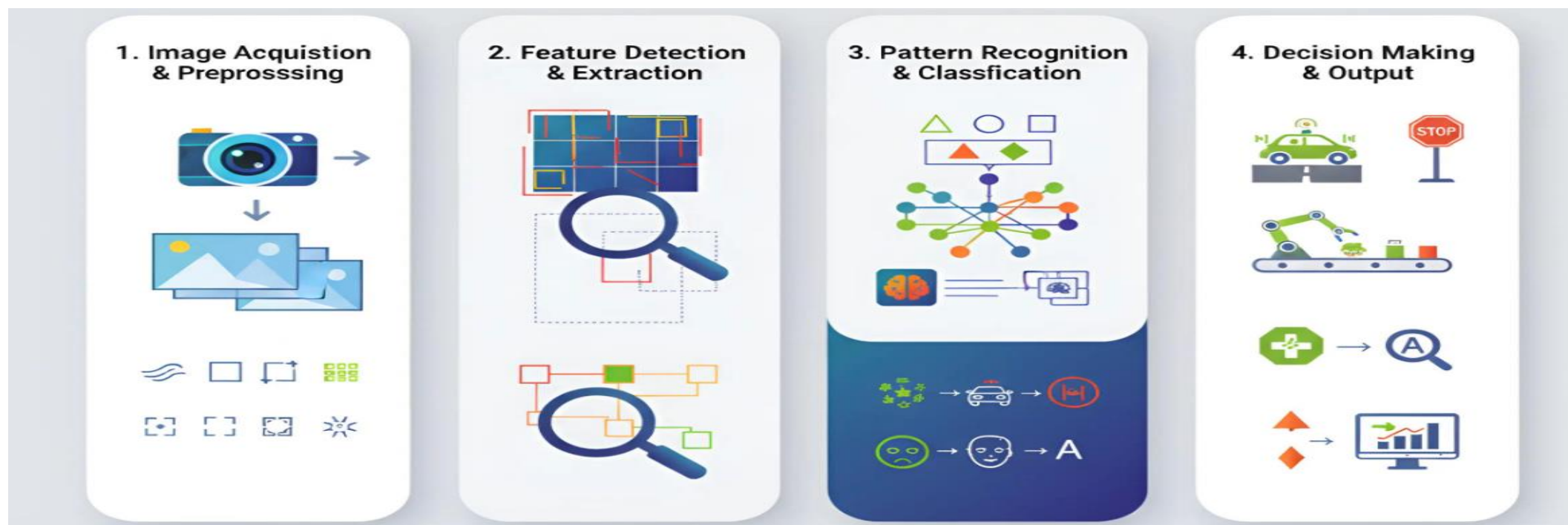


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Image Processing Pipeline



This slide explains the complete workflow. Every stage depends on the previous one. If image acquisition or sampling is poor, the computer vision system cannot make accurate decisions.



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Computer Vision Applications

Real-World Engineering Applications

Domain

Healthcare

Transportation

Manufacturing

Agriculture

Security

Robotics

Application

Tumor Detection, MRI Analysis

Autonomous Vehicles, Traffic Monitoring

Quality Inspection, Defect Detection

Crop Disease Detection

Face Recognition, CCTV Surveillance

Navigation and Object Detection



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- **Advantages**

- Improves image quality for better analysis.
- Reduces manual inspection and human errors.
- Handles large volumes of image data efficiently.
- Provides reliable and consistent results.
- Enhances decision-making using visual data.
- Works effectively in automation and robotics.

- **Challenges**

- Sensitive to poor lighting conditions.
- Image noise affects processing accuracy.
- Motion blur reduces image clarity.
- Low-resolution images lose important details.
- Requires high computational power.
- Environmental conditions can affect image quality
- Object occlusion makes recognition difficult.



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Conclusion

- Digital image fundamentals provide the foundation for representing images as pixels.
- Image acquisition captures real-world scenes using cameras and sensors. Image formation creates a focused image through optical lenses.
- Sampling converts the image into pixels, while quantization converts intensity values into digital levels.
- Computer Vision analyzes these digital images using AI algorithms to recognize objects, classify scenes, and make intelligent decisions.



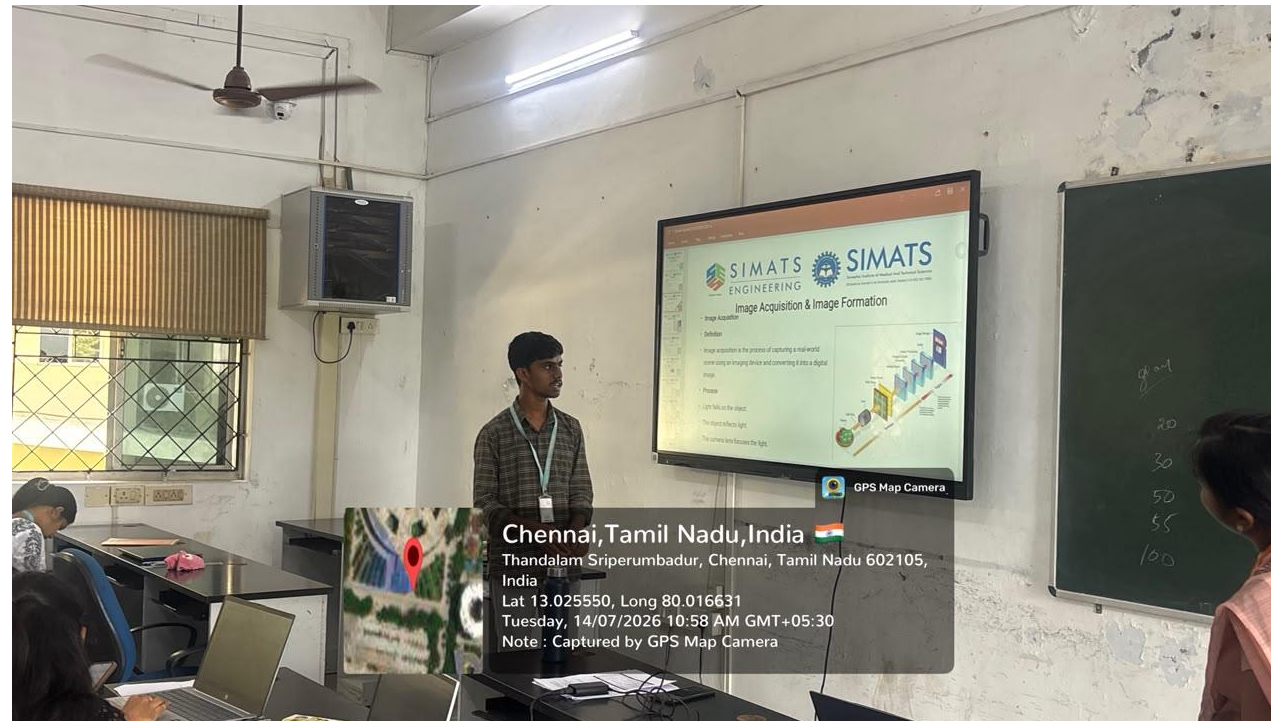
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Thank you